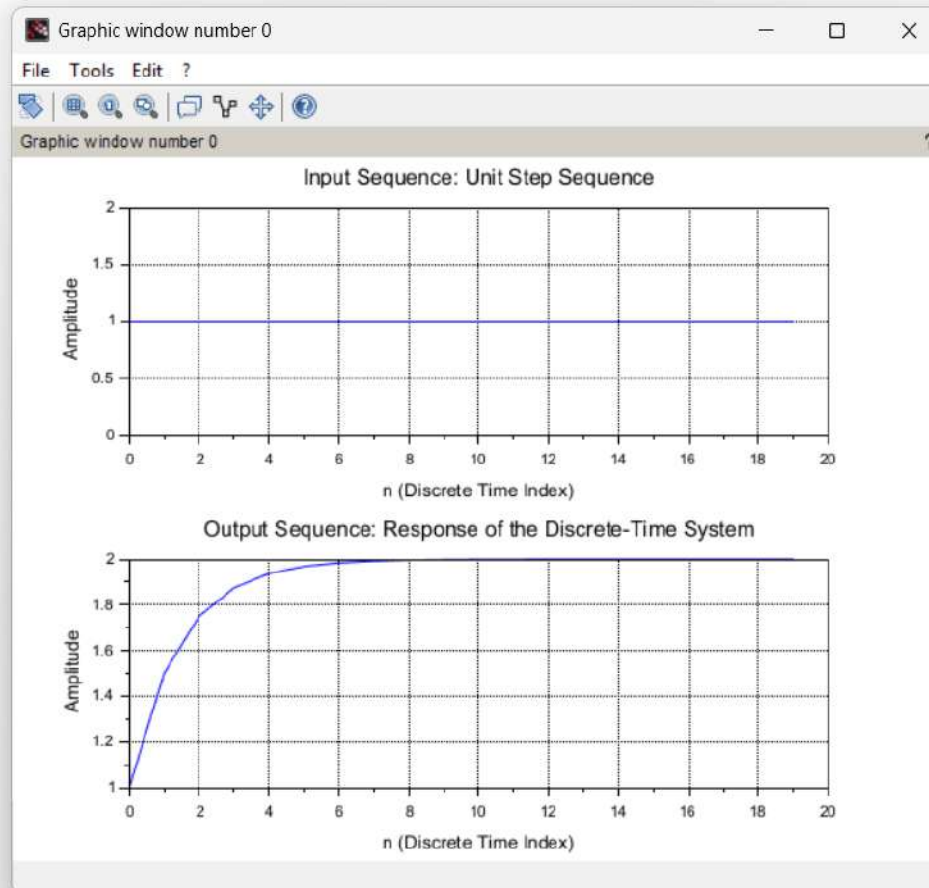


EXP_4.1 SS.sci (E:\EXP_4.1 SS.sci) - SciNotes

EXP_4.1 SS.sd

```
1 clc;
2 clear;
3 clf;
4
5 // Test Case: Analysis of a Discrete-Time Linear System
6
7 // Define parameters
8 N = 20;
9 n = 0:N-1;
10
11 // Input sequence (Unit Step)
12 x_n = ones(1,N);
13
14 // Output sequence initialisation
15 y_n = zeros(1,N);
16
17 // Initial condition
18 y_n(1) = x_n(1);
19
20 // Difference equation: y[n] = 0.5y[n-1] + x[n]
21 for k = 2:N
22     y_n(k) = 0.5 * y_n(k-1) + x_n(k);
23 end
24
25 // Plot Input Sequence
26 subplot(2,1,1);
27 plot(n, x_n);
28 xlabel("n (Discrete Time Index)");
29 ylabel("Amplitude");
30 title("Input Sequence: Unit Step Sequence");
31 xgrid();
32
33 // Plot Output Sequence
34 subplot(2,1,2);
35 plot(n, y_n);
36 xlabel("n (Discrete Time Index)");
37 ylabel("Amplitude");
38 title("Output Sequence: Response of the Discrete-Time System");
39 xgrid();
40
```



```

1  clc;
2  clear;
3  clf;
4
5  %% Test Case: Analysis of a Discrete-Time System
6
7  %% Define the input signal x[n]
8  n = 0:20;
9  x_n = (n >= 0) - (n >= 10); ... %% Rectangular pulse from n=0 to n=9
10
11 %% Initialize the output signal
12 y_n = zeros(1, length(n));
13
14 %% System coefficients
15 b = [0.5 0.3 0.2];
16
17 %% Compute the output
18 for i = 3:length(n)
19     y_n(i) = b(1)*x_n(i) + b(2)*x_n(i-1) + b(3)*x_n(i-2);
20 end
21
22 %% Plot input signal
23 subplot(2,1,1);
24 plot2d3(n, x_n);
25 xlabel("n - (Discrete Time Index)");
26 ylabel("x[n]");
27 title("Input Signal x[n]");
28
29 %% Plot output signal
30 subplot(2,1,2);
31 plot2d3(n, y_n);
32 xlabel("n - (Discrete Time Index)");
33 ylabel("y[n]");
34 title("Output Signal y[n] of the System");
35

```

